

Lab 3: Architectural Justification

Community Lost & Found Matching Network

Name: Niyatee Singh SRN: PES1UG24CS305

Architecture Selection

I chose **Microservices Architecture** for the Community Lost & Found Matching Network.

My system has a client app that people use to report lost or found items, plus four backend services (Item Management, Tag Extraction, Geo-Matching and Ownership Verification) and an Item Database. Each service runs on its own. The services talk to each other through REST APIs, and the database is accessed with SQL queries.

Why Not the Other Styles?

I looked at Layered and Client-Server first. Layered would have been the easiest to build, but tag extraction from photos is slow, and in a layered design it sits in the same business layer as everything else, so I couldn't scale just that part. Client-Server puts everything on one server, which means that if it goes down, nobody can report or claim anything. Microservices takes more work to set up and manage, but for this project I felt that was worth it.

Reasons for My Choice

1. **Some features need much more processing than others.** Reading tags out of an uploaded photo takes a lot more work than showing a list of items. Because Tag Extraction is its own service, I can run extra copies of it when lots of people are uploading, without touching the rest of the system.
2. **Features change separately and fail separately.** I expect to keep adjusting how matching works, for example how close two locations have to be to count as a match. Since Geo-Matching is a separate service, I can update it without redeploying everything else. And if one service crashes, like Tag Extraction, people can still report items and verify ownership.

Security Advantage

Only the Ownership Verification Service can see the security questions and answers. When someone tries to claim an item, the client sends their answers to this service and only gets back whether they passed or not. So even if another service were attacked, the answers wouldn't be exposed, and nobody could use them to claim items that aren't theirs. All traffic from the client also goes over HTTPS.

Performance Benefit

When a user uploads a lost or found item, Item Management saves the report straight away and sends the photo off for tagging in the background. The user doesn't have to wait for tagging to finish, so uploading feels quick. Geo-Matching also has its own connection to the database for location searches, so it can be scaled up by itself as more items get reported.